AL-2001
Artificial Intelligence
Lab # 10

Objectives:

- Pandas

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. Do not copy from any source otherwise you will be penalized with negative marks.
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

Lab Task 1: Data Import and Exploration

1.1. Load the Titanic dataset into a Pandas DataFrame and check for any missing values.

1.2. Access the information of the passenger with the highest fare in the dataset.

1.3. Create a new DataFrame containing only passengers who survived.

1.4. Display the count of passengers by each ticket class.

1.5. Explore the distribution of ages among male and female passengers using a suitable visualization.

Lab Task 2: Data Cleaning and Preprocessing

2.1. Impute missing values in the "age" column using the mean age of passengers.

2.2. Convert the "embarked" column to a categorical data type.

2.3. Remove rows where both the "age" and "cabin" information are missing.

2.4. Create a new column, "Family_Size," by adding the "sibsp" and "parch" columns.

2.5. Filter the dataset to include only passengers who paid a fare above the mean fare.

Lab Task 3: Data Analysis and Visualization

3.1. Visualize the survival rate based on ticket class using a bar chart.

3.2. Calculate and plot the correlation matrix for numerical features in the dataset.

3.3. Create a histogram to show the distribution of ages among passengers.

3.4. Calculate the percentage of passengers who embarked from each port.

3.5. Explore the relationship between the number of siblings/spouses ("sibsp") and the number of parents/children ("parch") using a scatter plot.

Lab Task 4: Advanced Data Manipulation

4.1. Sort the DataFrame based on the "age" column in descending order.

4.2. Use the "groupby" function to find the average fare paid by passengers in each ticket class.

4.3. Split the dataset into two DataFrames: one for passengers aged 18 and below and another for passengers above 18.

4.4. Create a new column, "Fare_Category," with values "Low," "Medium," and "High" based on the fare distribution.

4.5. Calculate the total number of passengers for each combination of ticket class and survival status.

Data Dictionary of Dataset

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Sex	
Age	Age in years	
sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	

ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

Variable Notes

pclass: A proxy for socio-economic status (SES)
1st = Upper
2nd = Middle
3rd = Lower

age: Age is fractional if less than 1. If the age is estimated, is it in the form of xx.5

sibsp: The dataset defines family relations in this way...
Sibling = brother, sister, stepbrother, stepsister
Spouse = husband, wife (mistresses and fiancés were ignored)

parch: The dataset defines family relations in this way...
Parent = mother, father
Child = daughter, son, stepdaughter, stepson
Some children travelled only with a nanny, therefore parch=0 for them

You need to do your exercise within the given time.